

HyperTrack: Hyperhidrosis Monitoring and Management System

Project Proposal for UCS503

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1 Higher-order goal

HyperTrack aims to improve the consistency and usefulness of hyperhidrosis symptom monitoring by reducing the manual effort required to record sweating episodes and understand patterns over time. The immediate engineering goal is to turn user-recorded, structured data into a reliable monitoring, analytics, and reporting workflow.



Figure 1: Problem context and motivation for structured monitoring.

2 Time-to-value

- Build the core record–store–analyze–report workflow first and add advanced ML features later.
- Deliver incremental functionality through short development iterations and continuous testing.
- Measure usefulness through recording completeness, dashboard functionality, report generation, and ML evaluation where implemented.

3 Problem Statement

Hyperhidrosis can involve excessive sweating of areas such as the palms, feet, underarms, and face. A practical challenge is the lack of consistent records covering severity, duration, affected area, activity, stress, and possible triggers. Without structured history, users may find it difficult to identify recurring patterns.

HyperTrack addresses this gap by allowing users to record episodes in a structured database, visualize historical trends, and generate an organized monitoring report. The system is a monitoring and decision-support tool; it does not diagnose or treat hyperhidrosis.

4 Objectives

- Provide a simple interface for recording sweating episodes and context.
- Store records securely in a structured relational database.
- Analyze frequency, severity, duration, affected areas, and possible triggers.
- Provide dashboards with statistics and historical trends.
- Support treatment/session tracking without issuing prescriptions.
- Generate an organized monitoring report.
- Investigate supervised ML for personalized severity prediction.
- Apply software engineering practices across analysis, design, development, and testing.

5 Proposed Solution

The proposed web application contains eight core modules:

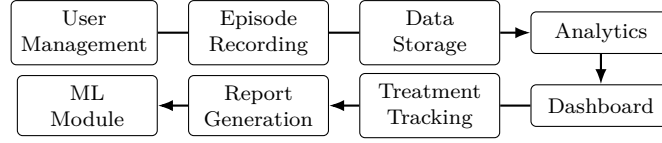


Figure 2: Core functional modules of HyperTrack.

6 System Architecture

The system follows a three-tier architecture. The presentation layer provides the interface, the application layer handles APIs, analytics and ML services, and the data layer stores users, episodes, triggers and treatment records.

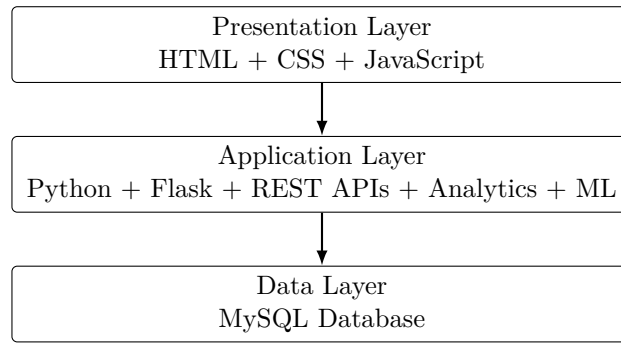


Figure 3: Three-tier system architecture.

7 System Workflow

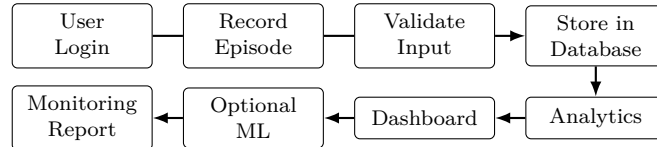


Figure 4: End-to-end operational workflow.

8 Episode Data Collection

Each episode records date/time, severity, duration, affected body area, activity, self-reported stress, possible trigger, temperature and humidity where available, and additional notes.

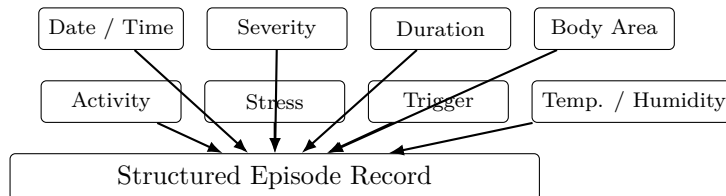


Figure 5: Structured data captured for each episode.

9 Data Processing and Analytics

Records are cleaned and processed to derive daily/weekly frequency, average severity, average duration, frequently affected areas, common possible triggers, severity trends, and associations between recorded stress and severity.

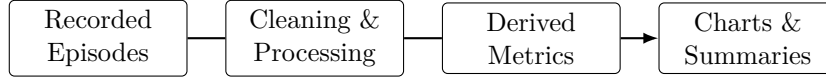


Figure 6: Analytics pipeline and derived metrics.

10 Machine Learning Component

The AIML component investigates supervised severity classification. Candidate features include stress, temperature, humidity, activity, time of day, previous severity, duration, caffeine intake, and affected body area. Candidate algorithms are Logistic Regression, Decision Trees, and Random Forest; the target is **Low / Moderate / High**.

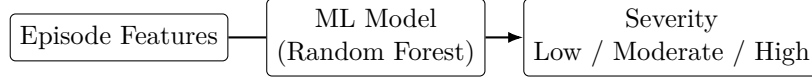


Figure 7: Proposed machine learning pipeline.

11 Testing and Validation

Testing is performed throughout development.

- **Unit Testing:** login, episode creation, validation and individual functions.
- **Integration Testing:** frontend, backend, database and analytics communication.
- **System Testing:** complete workflow from login to report generation.
- **User Acceptance Testing:** usability and clarity evaluation.
- **ML Evaluation:** accuracy, precision, recall, F1-score and confusion matrix, if ML is implemented.

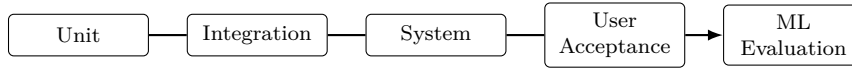


Figure 8: Multi-level testing and validation strategy.

12 Technology Stack

Component	Technology
Frontend	HTML, CSS, JavaScript
Backend	Python, Flask, REST API
Database	MySQL
Data Processing	Pandas, NumPy
Machine Learning	Scikit-learn
Visualization	Chart.js / Matplotlib
Testing	Python testing framework
Version Control	Git and GitHub
Development	VS Code / PyCharm



Figure 9: Technology stack across the application.

13 Development Timeline

Weeks	Task	Deliverable
1–2	Requirements and problem study	SRS
3–4	Architecture and database design	Architecture + ER/UML
5–6	Frontend and authentication	Working interface
7–8	Episode recording + database	Monitoring module
9–10	Dashboard + analytics	Charts and trends
11	Treatment tracking + reports	Report module
12	ML investigation	Prediction prototype
13	Integration + system testing	Test results
14	User acceptance + refinement	Validated prototype
15	Documentation + presentation	Final report

14 Resources and Budget

The project is developed by two students. Aditya Rajput focuses on backend development, database integration, ML, and system integration. Prabhkirat Kaur focuses on frontend development, UI/UX, documentation, testing, and analytics integration. Both participate in requirements, design, testing, and final documentation.

The project primarily uses freely available/open-source software: Python, Flask, MySQL, Pandas, NumPy, Scikit-learn, HTML/CSS/JavaScript, Git/GitHub, and VS Code/PyCharm. No external funding is required for the initial implementation.

15 Risks and Mitigation

- **Limited data:** use suitable public/synthetic data for the ML prototype and consider anonymized data in future.
- **Data privacy:** collect only necessary information and protect it through authentication and database security.
- **Medical interpretation:** clearly position the system as monitoring/analysis, not diagnosis or treatment.
- **ML reliability:** present predictions only as supplementary insights.
- **Development delays:** prioritize core monitoring before advanced ML.
- **User adoption:** keep episode recording quick and simple.

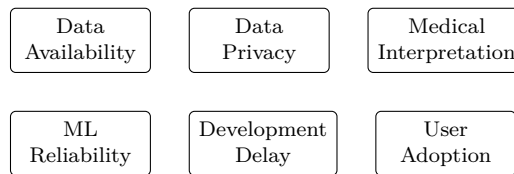


Figure 10: Key project risks.

16 Expected Outcomes

At completion, HyperTrack is expected to provide:

- structured episode records and historical views;
- frequency, severity and trigger trends;
- treatment/session history;

- organized monitoring reports; and
- optional ML-based personalized severity insights.

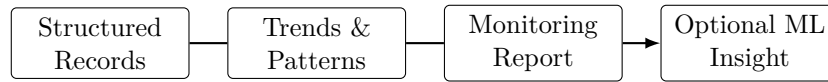


Figure 11: Expected project outcomes.

17 Summary

HyperTrack is a software-based monitoring and management system for organizing and analyzing hyperhidrosis-related sweating episodes. It combines structured data collection, database management, visualization, reporting, and an optional AIML component for severity prediction. The project emphasizes requirements engineering, modular architecture, testing, usability, privacy, and scalability.

Clinical Scope Note: HyperTrack is a monitoring and decision-support system. Its analytics and ML outputs are supplementary insights from user-recorded data and are not medical diagnosis, treatment recommendations, or medical advice.

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